

# String Amplitudes

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## 0.1 An attempt at Implementing the $i\varepsilon$ Prescription

Bianchi eqn (4.18)

$$\mathcal{A}^{(N)} = \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^{\frac{9}{2}} |\vartheta'_1(0|\tau)|^{4(N-1)}} \sum_{k=0}^{N-2} \binom{N-2}{k}^2 \sum_{n, \bar{n}=0}^{N-2} \binom{k}{n} \binom{k}{\bar{n}} \tilde{e}_1(\tau)^n \overline{\tilde{e}_1(\tau)}^{\bar{n}} f_1(\tau)^{k-n} \overline{f_1(\tau)}^{k-\bar{n}} \mathcal{I}_{2n+2, 2\bar{n}+2, 2N-4, 2N-4-2\bar{n}}(\tau, \bar{\tau}).$$

### 0.1.1 The Weierstrass Contribution $W(q, \bar{q})$

We define  $\tilde{e}_1(\tau) := e_1(\tau) - \mathcal{E}(\tau)$  and  $e_{s-1}$  are the Weierstrass invariants and  $\mathcal{E}(\tau)$  is a quasi-holomorphic modular form

$$e_{s-1}(\tau) = -4\pi i \partial_\tau \log \frac{\vartheta_s(0, \tau)}{\eta(\tau)}$$

$$\mathcal{E}(\tau) = 4\pi i \partial_\tau \log [\eta(\tau) \sqrt{\tau_2}].$$

Using these definitions

$$\begin{aligned} \tilde{e}_1(\tau) &= \left[ -4\pi i \partial_\tau \log \frac{\vartheta_s(0, \tau)}{\eta(\tau)} \right] - [4\pi i \partial_\tau \log [\eta(\tau) \sqrt{\tau_2}]] \\ &= -4\pi i \partial_\tau \left[ \ln \vartheta_2(0|\tau) - \ln \eta(\tau) + \ln \eta(\tau) + \frac{1}{2} \ln \tau_2 \right] \end{aligned}$$

We have  $\vartheta_2(0|\tau) = 2q^{\frac{1}{8}} \prod_{x \geq 1} (1 - q^x)(1 + q^x)^{2^1}$ .

$$\begin{aligned} \tilde{e}_1(\tau) &= -4\pi i \partial_\tau \left[ \ln 2 + \ln q^{\frac{1}{8}} + \sum_{x \geq 1} \ln(1 - q^x) + \sum_{x \geq 1} \ln(1 + q^x) + \frac{1}{2} \ln \tau_2 \right] \\ \tilde{e}_1(\tau) &= \pi^2 - \frac{\pi}{\tau_2} + 8\pi^2 \sum_{x \geq 1} x q^x \left[ -\frac{1}{1 - q^x} + \frac{2}{1 + q^x} \right] \\ A(\tau) &:= 8\pi^2 \sum_{x \geq 1} x q^x \left[ -\frac{1}{1 - q^x} + \frac{2}{1 + q^x} \right] \quad B(\tau_2) := \pi^2 - \frac{\pi}{\tau_2}. \end{aligned}$$

Working with the holomorphic part we can use the geometric series

$$\begin{aligned} -\frac{1}{1 - q^x} + \frac{2}{1 + q^x} &= \sum_j -q^{xj} + 2(-q^x)^j \\ &= \sum_j q^{xj} (-1 + 2(-)^j). \end{aligned}$$

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<sup>1</sup>check power q out front

Thus, we have

$$A(\tau) = 8\pi^2 \sum_{x \geq 1} \sum_{j=0}^{\infty} x q^{(j+1)x} \left( 2(-)^j - 1 \right).$$

Which gives

$$\begin{aligned} \tilde{e}_1(\tau) &= 8\pi^2 \sum_{x \geq 1} \sum_{j=0}^{\infty} x q^{(j+1)x} \left( 2(-)^j - 1 \right) + \pi^2 - \frac{\pi}{\tau_2} \\ \tilde{e}_1(\tau) &= A(\tau) + B(\tau_2) \end{aligned}$$

In the integrand we have terms of the form  $\tilde{e}_1^n(\tau)$  thus we have nested binomials

$$\begin{aligned} W(q) &:= \tilde{e}_1^n(\tau) = \sum_{p=0}^n \binom{n}{p} A(\tau)^p B^{n-p} \\ B^{n-p} &= \sum_{l=0}^{n-p} \binom{n-p}{l} (\pi^2)^{(n-p-l)} (-\pi)^l \tau_2^{-l}. \end{aligned}$$

The holomorphic part of the Weierstrass contribution is

$$W(q) = \sum_{p=0}^n \sum_{l=0}^{n-p} \binom{n}{p} \binom{n-p}{l} \left( \sum_{x \geq 1} \sum_{j=0}^{\infty} 8\pi^2 x q^{(j+1)x} \left( 2(-)^j - 1 \right) \right)^p \left( (\pi^2)^{n-p-l} (-\pi)^l \tau_2^{-l} \right) \quad (1)$$

We can define

$$S(\tau) := 8\pi^2 \sum_{m \geq 1} c(m) q^m, \quad c(m) = \sum_{\substack{x \geq 1, j \geq 0 \\ (j+1)x = m}} x \left( 2(-1)^j - 1 \right) \quad (2)$$

$$S^p = (8\pi^2)^p \sum_{M \geq 0} \sum_{\substack{m_1 + \dots + m_p = M \\ m_i \geq 1}} \prod_{\varpi=1}^p c(m_{\varpi}) q^M \quad (3)$$

Define  $C_p(M) := \sum_{\substack{m_1 + \dots + m_p = M \\ m_i \geq 1}} \prod_{\varpi=1}^p c(m_{\varpi})$ . Then we have

$$S^p = (8\pi^2)^p \sum_{M \geq 0} C_p(M) q^M \quad (4)$$

Thus, explicitly we have the holomorphic contribution

$$W(q) = \sum_{p=0}^n \sum_{l=0}^{n-p} \binom{n}{p} \binom{n-p}{l} (8\pi^2)^p \sum_{M \geq 0} C_p(M) q^M \left( \pi^{2((n-p)-l)} (-\pi)^l \tau_2^{-l} \right).$$

The if we consider the conjugate we get a net contribution of

$$W(q, \bar{q}) = \sum_{p=0}^n \sum_{l=0}^{n-p} \sum_{\bar{p}=0}^{\bar{n}} \sum_{\bar{l}=0}^{\bar{n}-\bar{p}} \sum_{M, \bar{M} \geq 0} (8\pi^2)^{p+\bar{p}} \binom{n}{p} \binom{n-p}{l} \binom{\bar{n}}{\bar{p}} \binom{\bar{n}-\bar{p}}{\bar{l}} C_p(M) \bar{C}_{\bar{p}}(\bar{M}) \\ \left( \pi^{2((n-p)-l)} (-\pi)^l \tau_2^{-l} \right) \left( \pi^{2((\bar{n}-\bar{p})-\bar{l})} (-\pi)^{\bar{l}} \tau_2^{-\bar{l}} \right) q^M \bar{q}^{\bar{M}}.$$

### 0.1.2 The vertex normalisation contribution $V(q, \bar{q})$ .

One may use the identity

$$\vartheta'_1(0|\tau) = 2\pi\eta^3(\tau) \quad \eta(\tau) := q^{\frac{1}{24}} \prod_{i=1}^{\infty} (1 - q^n) \quad (5)$$

So,

$$\vartheta'_1(0|\tau) = 2\pi q^{\frac{1}{8}} \prod_{i=1}^{\infty} (1 - q^n)^3 \\ |\vartheta'_1(0|\tau)|^{4(N-1)} = (2\pi)^{4(N-1)} q^{\frac{N-1}{4}} \bar{q}^{\frac{N-1}{4}} \prod_{i=1}^{\infty} (1 - q^n)^{6(N-1)} \prod_{i=1}^{\infty} (1 - \bar{q}^n)^{6(N-1)}.$$

If we define  $\hat{\eta}(\tau) := \prod_{i=1}^{\infty} (1 - q^n)^{6(N-1)}$ , then we have

$$|\vartheta'_1(0|\tau)|^{4(N-1)} = (2\pi)^{4(N-1)} q^{\frac{N-1}{4}} \bar{q}^{\frac{N-1}{4}} \hat{\eta}(\tau) \hat{\bar{\eta}}(\tau) \quad (6)$$

The contribution in the amplitude takes the form of inverse (6). We can expand  $\frac{1}{\hat{\eta}} = \sum_{j \geq 0} \tilde{c}_j q^j$ .

Thus, we have

$$\frac{1}{|\vartheta'_1(0|\tau)|^{4(N-1)}} = (2\pi)^{-4(N-1)} q^{-\frac{N-1}{4}} \bar{q}^{-\frac{N-1}{4}} \sum_{j \geq 0} \tilde{c}_j q^j \sum_{\bar{j} \geq 0} \tilde{\bar{c}}_{\bar{j}} \bar{q}^{\bar{j}}.$$

If we denote the vertex normalisation contribution (excluding the factor of  $2\pi$ )<sup>2</sup> by  $V(q, \bar{q})$  we may only consider the holomorphic part

$$V(q) = q^{-\frac{N-1}{4}} \sum_{j \geq 0} \tilde{c}_j q^j \quad (7)$$

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<sup>2</sup>remember this later XXX

We want to find a condition on the  $\tilde{c}_j$ . If we impose that <sup>3</sup>

$$\begin{aligned} \hat{\eta}(\tau) \cdot \frac{1}{\hat{\eta}(\tau)} &= 1 \quad \implies \left[ \sum_{r \geq 0} c_r^{\hat{\eta}} q^r \right] \left[ \sum_{j \geq 0} \tilde{c}_j q^j \right] = 1 \\ \left[ \sum_{r \geq 0} c_r^{\hat{\eta}} q^r \right] \left[ \sum_{j \geq 0} \tilde{c}_j q^j \right] &= \sum_{n \geq 0} \sum_{r=0}^n c_r^{\hat{\eta}} c_{n-r} q^n \\ \sum_{r=0}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} &= \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \geq 1 \end{cases} \\ &\implies c_0^{\hat{\eta}} \tilde{c}_n = 1. \end{aligned}$$

However, we know  $c_0^{\hat{\eta}}$  is the constant term of  $\hat{\eta}$  which by definition is 1. Thus,  $\tilde{c}_0 = 1$ .  
 $\forall n \geq 1$

$$\begin{aligned} \sum_{r=0}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} &= 0 \\ c_0^{\hat{\eta}} \tilde{c}_n + \sum_{r=1}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} &= 0 \\ \tilde{c}_n &= - \sum_{r=1}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} \quad \tilde{c}_0 = 1. \end{aligned}$$

So we have the holomorphic contribution

$$V(q) := q^{-\frac{N-1}{4}} \sum_{j \geq 0} \tilde{c}_j q^j \quad \text{with } \tilde{c}_0 = 1, \quad \tilde{c}_j = - \sum_{r=1}^j c_r^{\hat{\eta}} \tilde{c}_{j-r}.$$

and the net vertex normalisation contribution

$$V(q, \bar{q}) := \frac{1}{|\vartheta'_1(0|\tau)|^{4(N-1)}} = (2\pi)^{-4(N-1)} q^{-\frac{N-1}{4}} \bar{q}^{-\frac{\bar{N}-1}{4}} \sum_{j, \bar{j} \geq 0} \tilde{c}_j \tilde{c}_{\bar{j}} q^j \bar{q}^{\bar{j}}.$$

### 0.1.3 The $f_1$ Theta contribution

We have

$$f_1(\tau) := \left( \frac{\vartheta'_1(0|\tau)}{\vartheta_2(0|\tau)} \right)^2 = 8\pi^2 \left( \frac{1}{8} - q + 3q^2 - 4q^3 + \dots \right) \quad (8)$$

<sup>4</sup>. Thus we have

$$f_1(\tau) = \sum_{r \geq 0} b_r q^r \quad (9)$$

and in raising to a power we use the cauchy product

$$f_1(\tau)^{k-n} = \left( \sum_{r \geq 0} b_r q^r \right)^{k-n} = \sum_{i \geq 0} \left[ \sum_{r_1 + r_2 + \dots + r_{k-n} = i} b_{r_1} b_{r_2} \cdot b_{r_{k-n}} \right] q^i \quad (10)$$

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<sup>3</sup>-1 power?

<sup>4</sup>A.25 Bianchi

Define  $d_i^{k-n} := \sum_{r_1+r_2+\dots+r_{k-n}=i} b_{r_1} b_{r_2} \cdot b_{r_{k-n}}$ . Thus,

$$f_1(\tau)^{k-n} = \sum_{i \geq 0} d_i^{k-n} q^i \quad (11)$$

#### 0.1.4 The Lattice contribution $L(q, \bar{q})$

We have

$$\mathcal{A}^{(N)} \sim \int_{\mathcal{F}} \frac{d^2 \tau}{\tau_2^{\frac{9}{2}}} |\vartheta'_1(0|\tau)|^{4(N-1)} \sum_k \dots \mathcal{I}_{2n+2, \dots, \dots}(\tau, \bar{\tau}) \quad (12)$$

with

$$\begin{aligned} \mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2}(\tau, \bar{\tau}) &:= \frac{(-)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{\tau_2(N-1)}} \sum_{h=1}^{2(N-1)} \sum_{\mathbf{K} \in \mathbb{Z}^{2(N-1)}} \sum_{\bar{\mathbf{K}} \in \mathbb{Z}^{2(N-1)}} \delta_{\mathbf{1}_{2(N-1)} \cdot \mathbf{K}, h} \delta_{\mathbf{1}_{2(N-1)} \cdot \bar{\mathbf{K}}, h} \\ &\quad q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4(N-1)}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \bar{q}^{\frac{1}{2} \bar{\mathbf{K}} \cdot \bar{\mathbf{K}} - \frac{h^2}{4(N-1)}} e^{-i\pi \bar{\mathbf{a}} \cdot \bar{\mathbf{K}}}. \end{aligned}$$

$$\mathcal{I}_{2n+2, \dots, \dots}(\tau, \bar{\tau}) \sim \tau_2^{-\frac{1}{2}} \cdot \text{lattice} \quad (13)$$

If we combine the factor of  $\tau_2^{-\frac{1}{2}}$  into the  $\tau_2^{\frac{9}{2}}$  we have a holomorphic part remaining. Thus, define,

$$\theta_h^{[N_1]}(q) := \sum_{\substack{k \in \mathbb{Z}^{2(N-1)}, \\ \mathbf{1} \cdot \mathbf{k} = h}} q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4(N-1)}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \quad (14)$$

Fixing  $N_1$  fixes  $a$  and  $h$ .

$$g_h^{[N_1]} = \sum_{\substack{k \in \mathbb{Z}^{2(N-1)}, \\ \mathbf{1} \cdot \mathbf{k} = h}}^{\infty} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \quad (15)$$

Define  $m_{\text{lat}} := \frac{1}{2} k \cdot k - \frac{h^2}{4(N-1)}$

$$\theta_h^{[N_1]}(q) = \sum_{m_{\text{lat}}} g_h^{N_1}(m_{\text{lat}}) q^{m_{\text{lat}}} \quad (16)$$

So the lattice part can be written as

$$\begin{aligned} L(q, \bar{q}) &= \frac{(-)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{(N-1)}} \sum_{h=1}^{2(N-1)} \theta_h^{[N_1]}(q) \bar{\theta}_h^{[N_1]}(\bar{q}) \\ L(q, \bar{q}) &= \frac{(-)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{(N-1)}} \sum_{h=1}^{2(N-1)} \sum_{m_{\text{lat}}, \bar{m}_{\text{lat}}} g_h^{N_1}(m_{\text{lat}}) \bar{g}_h^{\bar{N}_1}(\bar{m}_{\text{lat}}) q^{m_{\text{lat}}} \bar{q}^{\bar{m}_{\text{lat}}}. \end{aligned}$$

where  $N_1 = 2n + 2$  and  $\bar{N}_1 = 2\bar{n} + 2$ .

## 0.2 The work on the board

MMM

$$\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} = \int_0^1 dx \int_0^1 dy e^{-4\pi(N-1)\tau_2 y^2} \theta_1(x + \tau y | \tau)^{N_1} \overline{\theta_1(x + \tau y | \tau)}^{\bar{N}_1} \theta_2(x + \tau y | \tau)^{N_2} \overline{\theta_2(x + \tau y | \tau)}^{\bar{N}_2}.$$

We have  $k^2 = \sum_i k_i^2$ , and  $k = (k_1, \dots, k_{2(N-1)})$ . Thus,

$$\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} = \int_0^1 dx \int_0^1 dy e^{-4\pi(N-1)\tau_2 y^2} \sum_{k, \bar{k} \in (\mathbb{Z} + \frac{1}{2})^{2(N-1)}} q^{\frac{1}{2}k^2} e^{2\pi i \sum_i k_i(x + \tau y)} (-1)^{\sum_i (k_i - \frac{1}{2})a_i} \bar{q}^{\frac{1}{2}\bar{k}^2} e^{-2\pi i \sum_i \bar{k}_i(x + \bar{\tau} y)} (-1)^{\sum_i (\bar{k}_i - \frac{1}{2})\bar{a}_i}.$$

With the constraints

$$\sum_{i=1}^{2(N-1)} k_i = \sum_{i=1}^{2(N-1)} \bar{k}_i = H \in \mathbb{Z} = h + M_{2(N-1)}, \quad h \in \{0, \dots, 2N-3\}, \quad M \in \mathbb{Z} \quad (17)$$

If we use the notation of Bianchi

$$k_i = l + k_i \quad \bar{k}_i = l + \bar{k}_i \quad (18)$$

So,

$$\begin{aligned} \mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \sum_{\sum_i k_i = \sum_i \bar{k}_i = H} \int_0^1 dy e^{-4\pi(N-1)\tau_2 y^2} q^{\frac{1}{2}k^2} e^{-4\pi\tau_2 y} (-1)^{\sum_i (K_i - \frac{1}{2})a_i} \\ &\quad \bar{q}^{\frac{1}{2}\bar{k}^2} e^{-2\pi i \bar{\tau} y H} (-1)^{\sum_i (\bar{k}_i - \frac{1}{2})\bar{a}_i} \\ &= \sum_M \sum_{h=0}^{2N-3} \sum_{\substack{\sum_{i=1}^{2(N-1)} k_i = \sum_{i=1}^{2(N-1)} \bar{k}_i \\ = h + 2M(N-1)}} \int_0^1 dy e^{-4\pi(N-1)\tau_2 (y+M)^2 + 4\pi(N-1)\tau_2 M^2 - 4\pi\tau_2 y h} \\ &\quad q^{\frac{1}{2}k^2} \bar{q}^{\frac{1}{2}\bar{k}^2} (-1)^{\sum_i (k_i - \frac{1}{2})a_i + \sum_i (\bar{k}_i - \frac{1}{2})\bar{a}_i}. \end{aligned}$$



which gives,

$$\begin{aligned}
\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \sum_{\substack{\sum_i k_i = \sum_i \bar{k}_i = h}} q^{\frac{1}{2} \sum_i k_i^2} \bar{q}^{\frac{1}{2} \sum_i \bar{k}_i} (-1)^{\sum_i (k_i - \frac{1}{2}) a_i + \sum_i (\bar{k}_i - \frac{1}{2}) \bar{a}_i} \int_{-\infty}^{\infty} dy e^{-4\pi(N-1)\tau_2 y^2 - 4\pi\tau_2 y h} \\
\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \sum_{\substack{\sum_i k_i = \sum_i \bar{k}_i = h}} q^{\frac{1}{2} \sum_i k_i^2} \bar{q}^{\frac{1}{2} \sum_i \bar{k}_i} (-1)^{\sum_i (k_i - \frac{1}{2}) a_i + \sum_i (\bar{k}_i - \frac{1}{2}) \bar{a}_i} \underbrace{\int_{-\infty}^{\infty} dy e^{-4\pi(N-1)\tau_2 \left(y + \frac{h}{N-1}\right)^2}}_{\frac{1}{\sqrt{(N-1)\tau_2}}} e^{\frac{4\pi}{N-1}\tau_1 h^2} \\
&= \frac{1}{\sqrt{(N-1)\tau_2}} \Theta_{A_{2N-3h}} (-1)^{\sum_i (k_i - \frac{1}{2}) a_i + \sum_i (\bar{k}_i - \frac{1}{2}) \bar{a}_i}
\end{aligned}$$

### 0.2.1 Collecting the contributions

$$\begin{aligned}
\mathcal{A}^{(N)} &= \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^{\frac{9}{2}} |\vartheta'_1(0|\tau)|^{4(N-1)}} \sum_{k=0}^{N-2} \binom{N-2}{k}^2 \sum_{n, \bar{n}=0}^{N-2} \binom{k}{n} \binom{k}{\bar{n}} \\
&\quad \tilde{e}_1(\tau)^n \bar{\tilde{e}}_1(\tau)^{\bar{n}} f_1(\tau)^{k-n} \bar{f}_1(\tau)^{k-\bar{n}} \mathcal{I}_{2n+2, 2\bar{n}+2, 2N-4, 2N-4-2\bar{n}}(\tau, \bar{\tau}).
\end{aligned}$$

Inspecting the powers of  $q$  we can define  $m := -\frac{N-1}{4} + j + M + i + m_{\text{lat}}$ . Likewise, for  $\tau_2$ , define  $\ell = l + \bar{l}$ .

$$F_{\ell}(m, \bar{m}) := \sum_{\substack{k, n, \bar{n} \\ p, \bar{p}, l + \bar{l} = \ell}} \binom{N-2}{k}^2 \binom{k}{n} \binom{k}{\bar{n}} \binom{n}{p} \binom{n-p}{l} \binom{\bar{n}}{\bar{p}} \binom{\bar{n}-\bar{p}}{\bar{l}} (8\pi^2)^{p+\bar{p}} \pi^{2(n-p-l)+2(\bar{n}-\bar{p}-\bar{l})} (-\pi)^{\ell} \quad (19)$$

$$\frac{(-1)^{n+\bar{n}}}{2\sqrt{N-1}} \sum_{h=1}^{2(N-1)} \mathcal{B}_{k, n, p, l}^{(h)}(m) \mathcal{B}_{k, \bar{n}, \bar{p}, \bar{l}}^{(h)}(\bar{m}) \quad (20)$$

where

$$\mathcal{B}_{k, n, p, l}^{(h)}(m) = \sum_{\substack{j, M, i, m_{\text{lat}} \geq 0 \\ -\frac{N-1}{4}, j+M+i+m_{\text{lat}}=m}} \tilde{c}_j C_p(M) d_i^{(k-n)} g_h^{2n+2}(m_{\text{lat}}).$$

Then we have

$$\mathcal{A}^{(N)} = (2\pi)^{-4(N-1)} \int_{\mathcal{F}} d^2\tau \tau_2^{-\frac{9}{2}} \sum_{\ell=0}^{2(N-1)} \tau_2^{-\ell} \sum_{m, \bar{m}} F_{\ell}(m, \bar{m}) q^m \bar{q}^{\bar{m}}.$$

We can split the integration region  $\mathcal{F} = \mathcal{F}_1 \cup R_\infty$ , where  $\mathcal{F}_1$  corresponds to  $\tau_2 \leq 1$  and  $R_\infty$  corresponds to  $-\frac{1}{2} \geq \tau_1 \leq \frac{1}{2}$  while  $\tau_2 \geq 1$ . This gives us

$$\mathcal{A}^{(N)} = \mathcal{A}_{\mathcal{F}_1}^{(N)} + \mathcal{A}_{\mathcal{F}_{Y-1}}^{(N)} \quad (21)$$

In terms of  $\tau_1$  and  $\tau_2$ ,

$$\mathcal{A}_{\mathcal{F}_{Y-1}}^{(N)} = (2\pi)^{-4(N-1)} \sum_{m, \bar{m}} \int_1^\infty d\tau_2 \tau_2^{-\frac{9}{2}} \sum_\ell F_\ell(m, \bar{m}) \tau_2^\ell e^{-2\pi\tau_2(m+\bar{m})} \int_{-\frac{1}{2}}^{\frac{1}{2}} d\tau_1 e^{2\pi i \tau_1(m-\bar{m})} \quad (22)$$

Considering the  $\tau_1$  contribution

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} d\tau_1 e^{2\pi i \tau_1(m-\bar{m})} = \frac{e^{\pi i(m-\bar{m})} - e^{-\pi i(m-\bar{m})}}{2\pi i(m-\bar{m})} = \frac{\sin[\pi(m-\bar{m})]}{\pi(m-\bar{m})} \quad (23)$$

Since  $m, \bar{m} \in \mathbb{Z}$  then,

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} d\tau_1 e^{2\pi i \tau_1(m-\bar{m})} = \delta_{m, \bar{m}} \quad (24)$$

Applying the delta function gives,

$$\mathcal{A}_{\mathcal{F}_{Y-1}}^{(N)} = (2\pi)^{-4(N-1)} \sum_m \int_1^\infty d\tau_2 \tau_2^{-\frac{9}{2}} \sum_\ell F_\ell(m, m) \tau_2^\ell e^{-4\pi\tau_2 m} \quad (25)$$